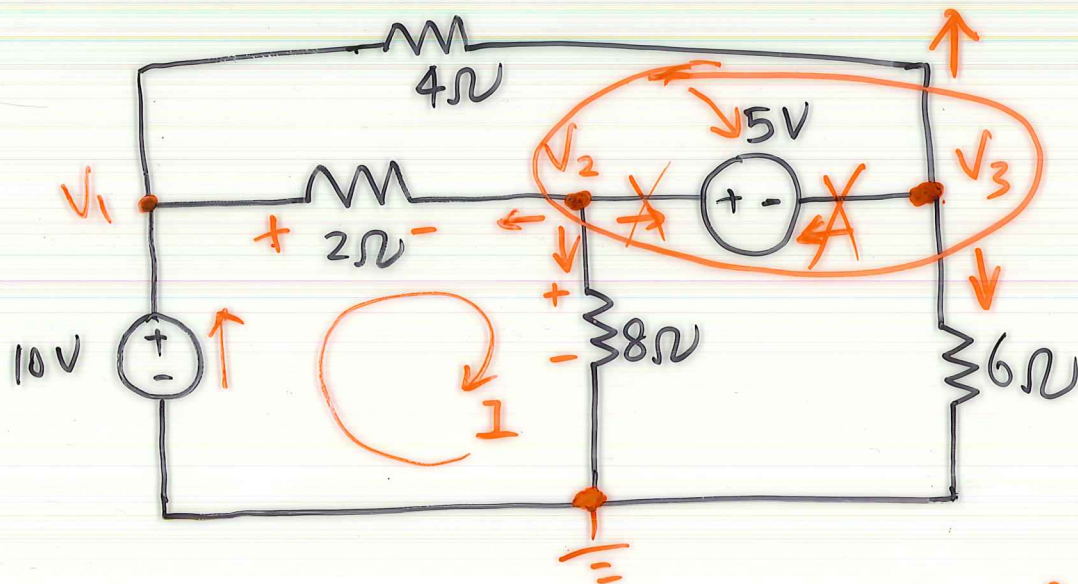


Supernode analysis

Def: Dependent / Independent voltage source connected between two non-ref node.



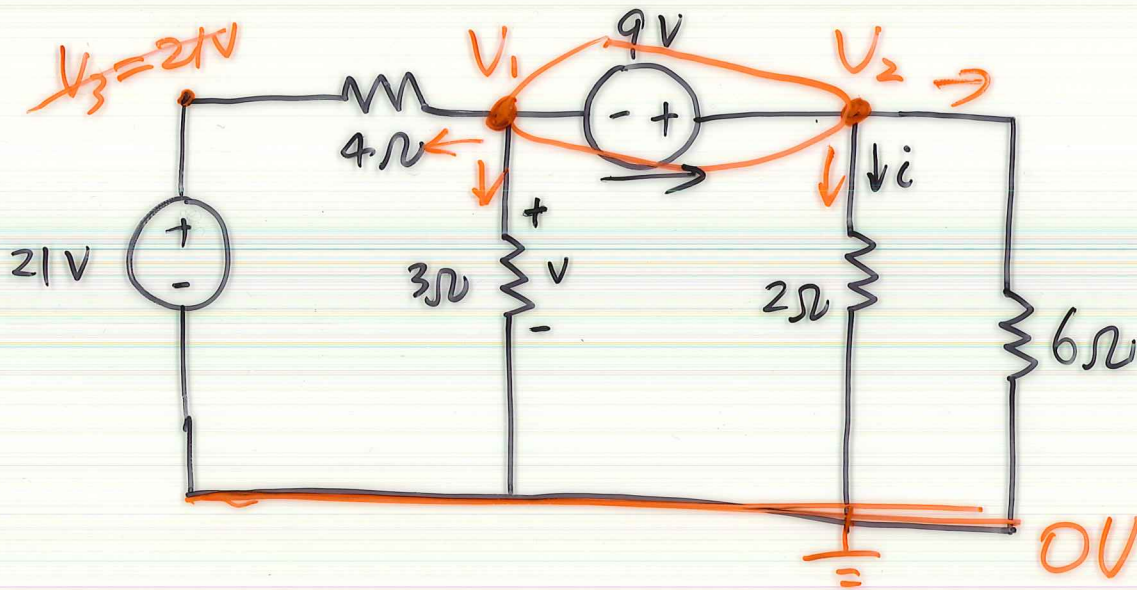
$$\underline{V_1 = 10V.}$$

$$\textcircled{1} V_2 - 5 = V_3$$

$$\begin{cases} -10 + 2I + 8I = 0 \\ 10 + (-2I) + (-8I) = 0. \end{cases}$$

$$\textcircled{2} \frac{V_2 - 10}{2} + \frac{V_2}{8} + \frac{V_3}{6} + \frac{V_3 - 10}{4} = 0$$

Find v and i .

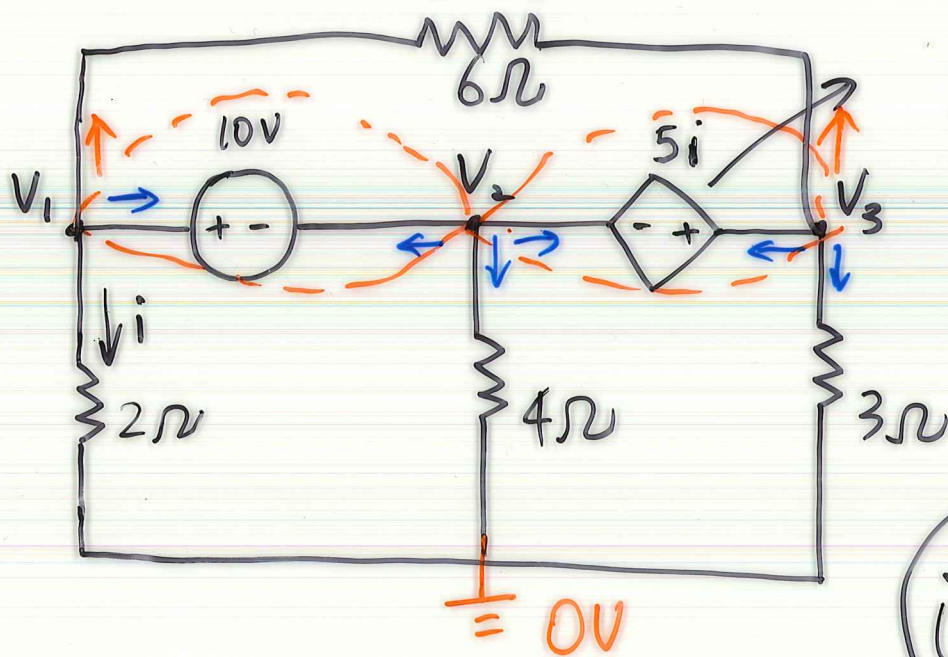


Eq 1: $V_1 + 9 = V_2$ ①

Eq 2: $\frac{V_1 - 21}{4} + \frac{V_1}{3} + \frac{V_2}{2} + \frac{V_2}{6} = 0$ ②

$V_1 = -0.6V, \quad V_2 = 8.4V.$

Find V_1 , V_2 , V_3 and i



CCVS

$$i = \frac{V_1}{2}$$

Eq 1: $V_1 - 10 = V_2$

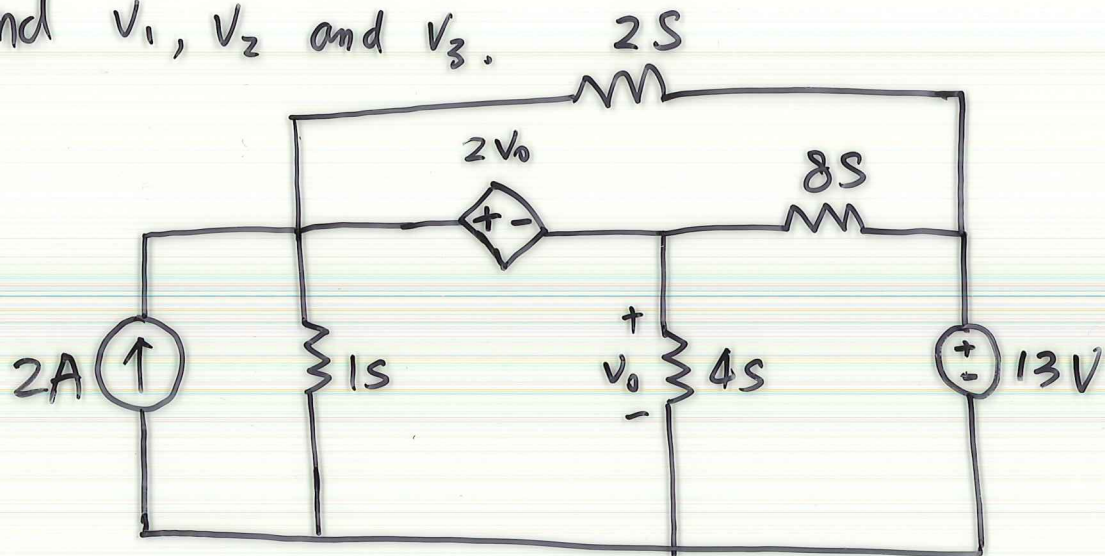
Eq 2: $V_2 + 5i = V_3$

Eq 3: $\frac{V_1}{2} + \frac{V_2}{4} + \frac{V_3}{3} = 0$ → to improve

Eq 3': $\frac{V_1 - V_3}{6} + \frac{V_1}{2} + \frac{V_2}{4} + \frac{V_3}{3} + \frac{V_3 - V_1}{6} = 0$

$V_1 = 3.043V$, $V_2 = -6.957V$
 $V_3 = 0.651V$

Find v_1 , v_2 and v_3 .



Find I_1, I_2, I_3, I_4

